ACADEMIC PROJECT REVIEW REPORT

Tri-Modal Reliability-Aware Clinical Decision Support System

"Can Uncertainty Warn Us? Cohort-Shift Reliability of Parkinson's Disease Classifiers Across Voice, Handwriting, and Gait"

Author: Mathesh

Domain: Medical AI / Uncertainty Quantification / Multimodal Learning

Modalities: Gait (VGRF) • Voice (Audio) • Handwriting (Spiral/Meander)

Technologies: PyTorch 2.0+ | Wav2Vec2 | ViT | Evidential Deep Learning

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Repository: github.com/Mathesh07/parkinson-s-disease-detection

RESEARCH PROTOTYPE ONLY — NOT A CERTIFIED MEDICAL DIAGNOSTIC DEVICE

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Abstract

Standard deep-learning classifiers in medical diagnostics output uncalibrated, overconfident softmax probabilities. When exposed to sensor degradation, acoustic noise, or cohort distribution shifts — differing clinical sites, acquisition hardware, or patient demographics — models frequently make catastrophic, high-confidence errors. This project implements an end-to-end, reliability-aware, tri-modal clinical decision support system for Parkinson's Disease (PD) detection from three non-invasive physiological biomarkers: Gait dynamics (Vertical Ground Reaction Force, VGRF), Vocal acoustics (16 kHz WAV audio), and Handwriting kinematic drawings (spiral and meander patterns). The central technical contribution is the integration of Evidential Deep Learning (EDL) — grounded in Dirichlet Subjective Logic — with Split Conformal Prediction, Deep Ensembles (M=5), Monte Carlo Dropout, and post-hoc Temperature Scaling across all three modalities, enabling each model to express calibrated, actionable uncertainty rather than mere softmax confidence. A novel five-case Cross-Modal Reliability Fusion Engine aggregates the uncertainty-weighted decisions from independent unimodal models into a final ACCEPT or FLAG verdict, explicitly handling missing modalities, cross-modal conflicts, and all-modality uncertainty. The system is benchmarked using Leave-One-Cohort-Out (LOCO) evaluation across three independent clinical cohorts for the Gait modality, and standard subject-level train-validation-test splits for Voice and Handwriting. Importantly, as the three datasets originate from different clinical studies with no shared patient identifiers, the system operates via decision-level fusion of independent unimodal models rather than conventional early or intermediate feature-level multimodal fusion. All preprocessing decisions — normalization, imputation, and splits — rigorously enforce zero patient data leakage at the subject level. An interactive Streamlit demonstration application provides real-time tri-modal inference with uncertainty dashboards, attention visualizations, and clinical evidence summaries.

Keywords

Parkinson's Disease Detection · Evidential Deep Learning · Dirichlet Uncertainty · Conformal Prediction · Deep Ensembles · Monte Carlo Dropout · Multimodal Fusion · Cohort-Shift Reliability · 1D CNN-BiLSTM · Wav2Vec2 · Vision Transformer · Gait Analysis · Voice Biomarkers · Handwriting Kinematics · VGRF · Distribution Shift

1. Introduction

Parkinson's Disease (PD) is the second most prevalent progressive neurodegenerative disorder worldwide, affecting approximately 10 million people globally. The condition is characterised by dopaminergic neuronal loss in the substantia nigra, resulting in characteristic motor symptoms including bradykinesia, resting tremor, rigidity, and postural instability. Diagnosis currently relies upon clinical observation by a specialist neurologist, and there is no definitive biomarker test available in routine clinical practice.

Digital biomarkers — measurable physiological signals captured by consumer or clinical sensors — have attracted substantial research interest as cost-effective, objective, and non-invasive complements to neurological assessment. Three biomarker classes are of particular relevance to PD:

- Gait dynamics: Quantified via instrumented insole Vertical Ground Reaction Force (VGRF) sensors that capture bilateral foot pressure patterns during walking. PD patients exhibit characteristic gait abnormalities including reduced stride length, increased double-support time, and freeze-of-gait.
- Vocal acoustics: The laryngeal muscles are affected early in PD, producing detectable changes in sustained phonation, reading tasks, and spontaneous dialogue. Key features include jitter, shimmer, and harmonics-to-noise ratio deviations.
- Handwriting kinematics: Motor impairments manifest in characteristic spiral and meander drawing abnormalities, including micrographia, tremor artefacts, and velocity reductions, which can be captured from static image scans.

While individual unimodal classifiers for PD detection have been extensively studied, a critical and largely unaddressed gap exists: standard deep learning models output uncalibrated softmax probabilities, provide no principled uncertainty estimates, and fail silently under distribution shift without warning the clinician. This project directly addresses this gap by implementing a comprehensive Reliability-Aware Tri-Modal System.

2. Background and Motivation

2.1 The Overconfident AI Problem in Clinical Decision Support

Standard deep neural networks trained with the cross-entropy loss produce softmax probability outputs that are theoretically interpretable as class probabilities. However, extensive empirical research has demonstrated that these probabilities are systematically overconfident — they frequently predict near-100% class probability for inputs that lie far outside the training distribution. In a clinical setting, this overconfidence is dangerous: a model may output 'Parkinson's Disease: 97% confident' for a corrupted or out-of-distribution sensor recording, presenting a misleading signal to a clinician.

This phenomenon was identified in the landmark study by Guo et al. (2017), who demonstrated that modern neural networks are significantly miscalibrated, and is particularly acute under domain or distribution shift — precisely the conditions encountered in real clinical deployments across different hospitals, patient populations, sensor hardware, and recording protocols.

2.2 The Silent Confidence Phenomenon

A central empirical finding of this project — termed the Silent Confidence Phenomenon — is that as signal corruption increases and classification error rises, predictive uncertainty does NOT reliably increase. Deep

models frequently become MORE overconfident in their misclassifications under distribution shift. Nominal softmax confidence is therefore an unsafe and insufficient safeguard against distribution shift in medical AI applications.

2.3 Research Focus and Central Research Question

This project investigates the following central research question:

> "Can Uncertainty Warn Us? Cohort-Shift Reliability of Parkinson's Disease Classifiers Across Voice, Handwriting, and Gait"

3. Problem Statement

Despite the existence of multiple published PD detection systems, three critical problems remain inadequately addressed:

P1: Lack of Principled Uncertainty: Existing classifiers output uncalibrated softmax probabilities without distinguishing between cases where predictions are trustworthy versus uncertain. This is particularly dangerous in clinical decision support.

P2: Silent Failure Under Distribution Shift: Models trained on one clinical cohort or recording setup regularly fail silently when deployed on data from a different site, sensor, or patient population — without any increase in reported uncertainty.

P3: Missing Modality and Cross-Modal Conflict Handling: Real-world clinical deployments rarely provide all three biomarker modalities simultaneously. Existing fusion approaches lack mechanisms to handle missing modalities, modality conflicts, or cases where individual modalities are unreliable.

3.1 Research Gap

The specific gap addressed is the absence of a principled, deployable, uncertainty-aware multimodal framework that: (1) quantifies prediction reliability at inference time without requiring distribution labels; (2) handles missing modalities gracefully; (3) is validated under realistic cross-cohort distribution shift conditions; and (4) provides actionable, explainable reliability decisions rather than raw probabilities.

4. Research Objectives and Scope

4.1 Main Aim

To design, implement, and evaluate a reliability-aware tri-modal system for Parkinson's Disease detection that can quantify its own prediction confidence and explicitly flag unreliable predictions, enabling safer clinical decision support.

4.2 Specific Objectives

- O1: Implement modality-specific deep learning architectures (1D-CNN+BiLSTM, Wav2Vec2, ViT) for Gait, Voice, and Handwriting PD classification.
- O2: Integrate Evidential Deep Learning (EDL) with Dirichlet Subjective Logic as the primary uncertainty quantification paradigm across all three modalities.
- O3: Benchmark and compare six uncertainty quantification methods: EDL, Deep Ensembles (M=5), MC Dropout (N=30), Conformal Prediction, Temperature Scaling, and softmax baseline.
- O4: Evaluate model reliability under controlled signal corruption and Leave-One-Cohort-Out (LOCO) distribution shift.
- O5: Implement a decision-level Cross-Modal Reliability Fusion Engine with five cases (A–E) covering all multi-modality and missing-modality scenarios.
- O6: Enforce strict subject-level data partitioning with zero patient leakage across all splits.
- O7: Build an interactive Streamlit demonstration application for real-time tri-modal clinical inference.

4.3 Project Scope and Constraints

- Scope: Binary classification (Parkinson's Disease vs. Healthy Control) from non-invasive digital biomarkers.
- Constraint: The three datasets originate from independent clinical studies with no shared patient identifiers. Cross-modal fusion operates at the decision level, not the feature level.
- Constraint: All experiments are designed for CPU execution (GPU acceleration is supported but not required).
- Constraint: The system is an academic research prototype — not a certified medical diagnostic device.
- Scope: Uncertainty evaluation under signal corruption and cohort shift, not under pathological heterogeneity or longitudinal progression.

5. Literature Review and Related Work

5.1 Parkinson's Disease Digital Biomarkers

Gait abnormalities in PD were quantified early using Vertical Ground Reaction Force (VGRF) sensors. Hausdorff et al. (2000) demonstrated that gait variability measured by VGRF correlates significantly with disease severity and fall risk. The PhysioNet Gait in Parkinson's Disease database (Goldberger et al., 2000; Hausdorff et al., 2007) aggregated data from three independent clinical studies (Ga, Ju, Si cohorts), providing the benchmark dataset widely used in gait-based PD classification.

Voice biomarkers for PD have been studied extensively since Little et al. (2009) demonstrated that dysphonia features extracted from sustained vowel phonation classify PD with high accuracy. More recently, Jaeger et al. (2019) collected the MDVR-KCL (King's College London) speech dataset using both read text and spontaneous dialogue tasks, enabling more ecologically valid evaluation of voice-based PD detection.

Handwriting analysis for PD was systematically evaluated by Pereira et al. (2016, 2018), who introduced the NewHandPD dataset containing spiral and meander drawing images from PD patients and healthy controls. They demonstrated that kinematic features from spiral drawings achieve competitive classification performance without requiring specialized drawing tablets.

5.2 Uncertainty Quantification in Deep Learning

Sensoy et al. (2018) [NeurIPS 2018]: Introduced Evidential Deep Learning (EDL), parameterising a Dirichlet distribution over class probabilities using Subjective Logic. The Expected Cross-Entropy (ACE) loss with KL divergence annealing trains networks to accumulate evidence rather than optimise softmax probabilities. This enables closed-form uncertainty decomposition into aleatoric and epistemic components.

Han et al. (2021) [ICLR 2021]: Proposed Trusted Multi-View Classification (TMC) extending EDL to multiple views, combining evidence across independent views using the Dempster-Shafer combination rule. This provides the theoretical foundation for multi-source evidence combination.

Lakshminarayanan et al. (2017) [NeurIPS 2017]: Deep Ensembles: training M=5 independent models with different random seeds produces calibrated uncertainty through ensemble disagreement. Demonstrated superior calibration compared to MC Dropout on out-of-distribution detection.

Gal & Ghahramani (2016) [ICML 2016]: Monte Carlo Dropout: activating dropout at test time enables approximate Bayesian inference. N=30 forward passes estimate predictive entropy and mutual information as epistemic uncertainty measures.

Angelopoulos & Bates (2021) [Statistical Science]: Split Conformal Prediction: provides finite-sample marginal coverage guarantees (1-α) without distributional assumptions beyond exchangeability. Non-conformity scores calibrated on a held-out calibration set determine prediction set inclusion thresholds.

Guo et al. (2017) [ICML 2017]: Demonstrated that modern neural networks are significantly miscalibrated despite high accuracy. Temperature Scaling — optimising a single scalar T on validation logits — is the most effective post-hoc calibration method.

5.3 Multimodal PD Detection Systems

Most published multimodal PD detection systems employ feature concatenation from separately collected datasets, which is methodologically valid only if all features can be assumed to represent independent measurements of the same underlying condition, not paired patient recordings. Tsanas et al. (2012) combined telemonitoring voice features with motor UPDRS scores. Sakar et al. (2019) combined voice and clinical features from the same patients. However, the combination of gait, voice, and handwriting data from different independent clinical studies — as in the present project — requires decision-level rather than feature-level fusion, as explored in this work.

6. Existing Systems, Limitations, and Proposed Approach

6.1 Limitations of Existing Approaches

Softmax Overconfidence: Virtually all existing PD classifiers use standard cross-entropy + softmax outputs. These are known to be overconfident and do not reflect true predictive uncertainty.

No Distribution Shift Evaluation: Most published works evaluate on random train-test splits from the same dataset. Cross-site or cross-cohort generalization is rarely assessed.

No Missing Modality Handling: Existing multimodal approaches assume all modalities are always available. Real clinical settings frequently lack one or more biomarker measurements.

No Conformal Coverage Guarantee: No published PD multimodal system provides statistically rigorous conformal prediction sets with finite-sample coverage guarantees.

Silent Failure Risk: Without principled uncertainty quantification, models can fail silently — making high-confidence incorrect predictions without triggering review.

Feature-Level Fusion of Unpaired Data: Some works inappropriately apply feature-level fusion to independently collected datasets, which is statistically invalid unless samples represent the same individuals.

6.2 Comparison Table: Existing vs. Proposed Approach

Dimension	Typical Existing Systems	Our System
Uncertainty Method	Softmax confidence (uncalibrated)	EDL + Ensemble + MC Dropout + Conformal
Modality Count	1–2 modalities	3 modalities (Gait, Voice, Handwriting)
Fusion Level	Feature or early fusion	Decision-level (reliability-weighted)
Missing Modality Support	Not addressed	Explicit 5-Case Policy (A–E)
Distribution Shift Eval	Rarely assessed	LOCO + Corruption Suite (5 types, 5 levels)
Conformal Coverage	Not provided	Split Conformal (90% nominal coverage)
Subject-Level Splits	Often window/recording-level	Strict subject-level, zero leakage
Calibration Eval	Accuracy only	ECE, Brier Score, Reliability Diagrams
Data Pairing	Same-patient multimodal	Independent cohort, decision-level fusion
Clinical Safeguard	None	ACCEPT/FLAG decision engine
Interactive Demo	Rarely	Full Streamlit application

7. Datasets: Sources, Characteristics, and Modality Correspondence

7.1 Gait Dataset: PhysioNet Gait in Parkinson's Disease

Property	Details
Official Name	Gait in Parkinson's Disease Database v1.0.0
Source	PhysioNet.org — goldberger2000physiobank
Original Papers	Hausdorff et al. (2007); Goldberger et al. (2000)
Total Recordings	306 walking trials
Total Subjects	165 (93 PD, 72 Healthy Control)
Cohort Ga (Galit et al.)	47 subjects (29 PD, 18 HC), 113 recordings
Cohort Ju (Hausdorff et al.)	54 subjects (29 PD, 25 HC), 129 recordings
Cohort Si (Frenkel-Toledo et al.)	64 subjects (35 PD, 29 HC), 64 recordings
Signal Type	Vertical Ground Reaction Force (VGRF) in Newtons

Property	Details
Channels	16 (L1–L8 left foot + R1–R8 right foot)
Sampling Rate	100 Hz (100 samples/second)
File Format	Tab-separated .txt (19 columns: time, 16 VGRF, 2 totals)
Data Splits Used	70% train / 15% val / 15% test (subject-level stratified)
Train Subjects	115 | Val: 25 | Test: 25
Train Recordings	226 | Val: 40 | Test: 40
Test Windows	1870 windows × (500 timesteps, 16 channels)

7.2 Voice Dataset: MDVR-KCL Speech Corpus

Property	Details
Official Name	MDVR-KCL (Multidimensional Voice Research – King's College London)
Source	King's College London, September 2017
Reference	Jaeger et al. (2019)
Signal Type	Single-channel WAV audio at variable sampling rates
Tasks	ReadText (read passage) and SpontaneousDialogue
Classes	HC (Healthy Control) and PD (Parkinson's Disease)
Target Sampling Rate	16,000 Hz (polyphase resampled from source)
Chunking	10-second chunks, 2-second overlap, 8-second stride
Chunk Samples	160,000 samples per chunk
File Format	WAV (mono, float32 after normalisation)
Data Splits Used	70% train / 15% val / 15% test (subject-level stratified)
Subject Split Validation	6 test subjects (HC: 3, PD: 3) per reported evaluation

7.3 Handwriting Dataset: NewHandPD

Property	Details
Official Name	NewHandPD — PD and HC Handwriting Drawing Dataset
Source	Pereira et al., Brazilian Computer Society
References	Pereira et al. (2016, 2018)
Tasks	Spiral drawings and Meander (zigzag) drawings
Image Folders	HealthySpiral, PatientSpiral, HealthyMeander, PatientMeander
Metadata Files	NewSpiral.csv, NewMeander.csv (ID_PATIENT, CLASS_TYPE, IMAGE_NA
Class Labels	CLASS_TYPE=1 (Healthy), CLASS_TYPE=2 (Parkinson)
Image Format	Static RGB scan images (resized to 224×224 for models)

Property	Details
Patient-Level Split	StratifiedGroupKFold (10 folds), patient-level grouping
Patient Split	10% test fold + 10% validation fold; remainder training
Test Set (ViT Evaluation)	56 samples (TP=24, TN=24, FP=8, FN=0)

7.4 Dataset Correspondence and the Modality Availability Problem

A critical methodological consideration governs the multimodal integration strategy. The three datasets originate from entirely independent clinical studies conducted at different institutions, at different times, using different patient populations and protocols. There are NO shared patient identifiers between the Gait (PhysioNet), Voice (MDVR-KCL), and Handwriting (NewHandPD) datasets.

Dataset	Gait (PhysioNet)	Voice (MDVR-KCL)	Handwriting (NewHandPD)
Gait (PhysioNet)	—	No shared patients	No shared patients
Voice (MDVR-KCL)	No shared patients	—	No shared patients
Handwriting (NewHandPD)	No shared patients	No shared patients	—

This absence of patient correspondence means that conventional early fusion, intermediate feature-level fusion, or joint multimodal training — where a single model receives features from all three modalities simultaneously — is methodologically invalid for this project. Section 11 addresses this problem in detail.

8. Data Preprocessing Pipeline

8.1 Gait VGRF Preprocessing

File Discovery and Parsing:

All .txt files in the PhysioNet directory are discovered. Filenames are parsed using a regex (e.g., GaCo01_01.txt → study=Ga, group=Co, subject=GaCo01, walk=01, label=0). Files matching format.txt, demographics.txt, and SHA256SUMS.txt are excluded.

Column Selection (16-channel VGRF extraction):

From 19 raw columns (time, L1–L8, R1–R8, TotalL, TotalR), only columns 1–16 (L1–L8, R1–R8) are extracted. The time column (col 0) and total columns (cols 17–18) are discarded.

NaN and Inf Cleaning:

np.nan_to_num(arr, nan=0.0, posinf=0.0, neginf=0.0) is applied to handle any corrupted sensor readings.

Subject-Level Stratified Split:

Subjects (not windows) are split into Train/Val/Test with class stratification. 70%/15%/15% ratios. Assertions enforce zero-leakage: no subject appears in more than one split.

Channel-wise Z-Score Normalization (Training-Only Fit):

GaitNormalizer fits channel-wise mean μ_c and standard deviation σ_c exclusively on training recordings. Validation and test recordings are transformed using frozen training statistics. x■_{t,c} = (x_{t,c} – μ_c^train) / (σ_c^train + ε), where ε = 1e-6.

Temporal Windowing (500 steps, 50% overlap):

Continuous recordings are sliced into 5.0-second windows of 500 timesteps at 100 Hz with 250-step stride (50% overlap). Short recordings are zero-padded to window size. Final tensor shape per window: (500, 16).

8.2 Voice Audio Preprocessing

File Discovery and Metadata Building:

Audio files are discovered by scanning Voice/26-29_09_2017_KCL/{ReadText,SpontaneousDialogue}/{HC,PD}/ subdirectories. Subject IDs are extracted from filenames via regex (IDxxx pattern).

Mono Conversion and Amplitude Normalisation:

Multi-channel audio is averaged to mono. Integer formats (int16, int32) are normalised to float32 in [-1, 1] range (÷32768 and ÷2147483648 respectively).

Polyphase Resampling to 16 kHz:

All audio files are resampled to 16,000 Hz using scipy.signal.resample_poly with GCD-based up/down factors for high fidelity.

Chunking with Overlap:

Audio is sliced into 10-second chunks (160,000 samples) with 2-second overlap (128,000 sample stride). Short recordings are zero-padded to one chunk.

Wav2Vec2 Processor Normalisation:

Each chunk passes through the facebook/wav2vec2-base Wav2Vec2Processor, which applies mean-zero unit-variance normalisation expected by the pretrained model.

8.3 Handwriting Image Preprocessing

CSV Loading and Manifest Building:

NewSpiral.csv and NewMeander.csv are loaded and concatenated. IMAGE_NAME is resolved to actual file paths by searching HealthySpiral, PatientSpiral, HealthyMeander, and PatientMeander folders.

Missing Image Handling:

Images where the referenced file does not exist are flagged (image_exists=False) and excluded from model inputs.

Patient-Level Split (StratifiedGroupKFold):

StratifiedGroupKFold with n_splits=10 and RANDOM_SEED=42 assigns patients (ID_PATIENT) to train, validation, or test folds. No patient appears in more than one split.

Training Augmentation:

RandomResizedCrop(224, scale=(0.85, 1.0)), RandomRotation(±10°), RandomHorizontalFlip(p=0.5).

ImageNet Normalisation:

Resize to 224×224, ToTensor(), Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) applied to all splits. Training uses augmentation before normalisation; val/test do not.

9. Model Architectures and Theoretical Foundations

9.1 Gait Model: 1D-CNN + Bidirectional LSTM with Evidential Head

The Gait model is designed from scratch to process continuous multivariate VGRF time-series data. It combines a 1D convolutional feature extractor for local temporal dynamics with a bidirectional LSTM for long-range stride periodicity and temporal dependencies.

Layer / Block	Input	Output	Parameters
Input (VGRF Window)	(B, 500, 16)	—	—
Transpose	(B, 500, 16)	(B, 16, 500)	—
CNN Block 1: Conv1d(16→64, k=5), BN, ReLU, MaxPool(2)	(B, 16, 500)	(B, 64, 250)	~5,000
CNN Block 2: Conv1d(64→128, k=5), BN, ReLU, MaxPool(2)	(B, 64, 250)	(B, 128, 125)	~41,000
Transpose	(B, 128, 125)	(B, 125, 128)	—
2-Layer BiLSTM (hidden=128, bidir=True, dropout=0.3)	(B, 125, 128)	(B, 125, 256)	~1,050,000
Global Temporal Mean Pool	(B, 125, 256)	(B, 256)	—
Embedding Layer: Linear(256→128), BN, ReLU, Dropout(0.3)	(B, 256)	(B, 128)	~33,000
Evidential Head: Dropout(0.2), Linear(128→2), Softplus	(B, 128)	(B, 2) ≥ 0	~258

The Evidential Head applies a Softplus activation (ln(1 + exp(z))) to the linear output, ensuring all evidence values e_k ≥ 0 as required by Dirichlet Subjective Logic. The output is a 2-dimensional vector [e_HC, e_PD] representing evidence for Healthy Control and Parkinson's Disease respectively.

Hyperparameter	Value	Rationale
CNN Channels	[64, 128]	Progressive spatial feature expansion
CNN Kernel Size	5	5 timesteps = 50ms, covers single heel-strike
LSTM Hidden Size	128 per direction (256 total)	Sufficient capacity for stride periodicity
LSTM Layers	2 (bidirectional)	Forward and backward temporal context
Embedding Dim	128	Compact representation for downstream fusion
Dropout	0.3 (LSTM), 0.2 (head)	Regularisation against small medical datasets
Batch Size	32	Empirically stable for VGRF window data
Learning Rate	1e-3 (Stage A), differential (Stage B)	Two-stage fine-tuning strategy
Epochs	50 (baseline), 10+15 (evidential)	Early stopping with patience=10
Optimizer	AdamW (weight_decay=1e-4)	Decoupled weight decay for better generalisation
Max Grad Norm	1.0	Gradient clipping for LSTM stability
Random Seed	42	Full reproducibility

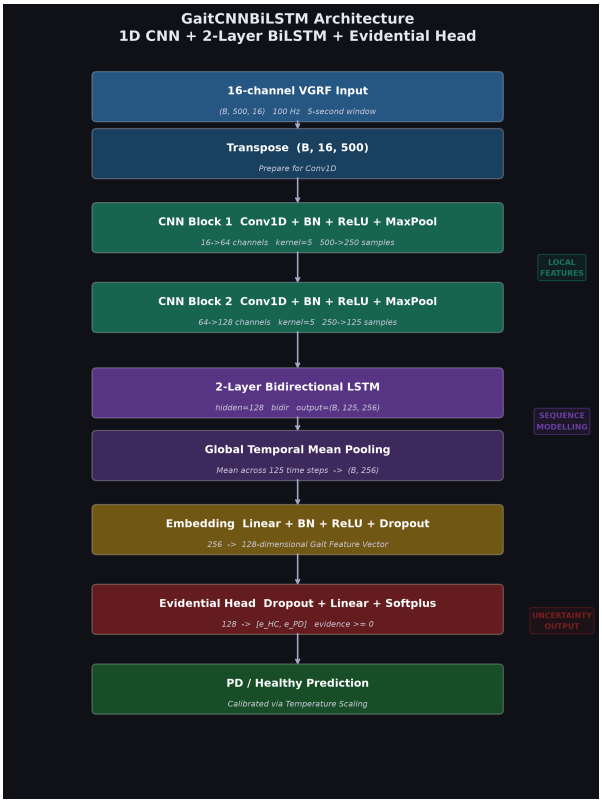


Figure 9.1: GaitCNNBiLSTM Architecture — 1D-CNN feature extraction, BiLSTM temporal modelling, and Evidential Head.

9.2 Voice Model: Wav2Vec2 with Evidential Head

The Voice modality leverages the self-supervised pretrained Wav2Vec2 model (facebook/wav2vec2-base) from Facebook Research / Meta AI. Wav2Vec2 learns powerful speech representations from large unlabelled audio corpora using a contrastive self-supervised objective, enabling strong transfer to downstream tasks with limited labelled data — precisely the scenario for medical voice datasets.

Component	Description	Output Dim
Input Audio	16 kHz WAV, 10-sec chunk (160,000 samples)	(B, 160000)
Wav2Vec2 Feature Encoder (frozen)	7-layer 1D-CNN, converts raw audio to latent vectors at 50 Hz	(B, T, 512)
Wav2Vec2 Transformer Encoder	12-layer Transformer, produces contextualised speech representations	(B, T, 768)
Masked Temporal Mean Pooling	Mean over time dimension (with attention mask)	(B, 768)
Evidential Head: Dropout(0.1), Linear(768→2), Softplus	Linear projection to evidence	(B, 2) ≥ 0

Hyperparameter	Value
Pretrained Backbone	facebook/wav2vec2-base (94M parameters)
Feature Encoder	Frozen during Stage A training
Batch Size	2 (memory constraint for 160k sample chunks)
Gradient Accumulation Steps	2 (effective batch size = 4)
Learning Rate (Stage A, head only)	1e-3
Learning Rate (Stage B, full model)	1e-5

Hyperparameter	Value
Weight Decay	0.01
Epochs	15 with patience=5
Max Grad Norm	1.0
Annealing Epochs (KL)	10
Random Seed	42

9.3 Handwriting Model: Vision Transformer (ViT) and CNN Baseline

The Handwriting modality employs a pretrained Vision Transformer (ViT-Base-Patch16-224-in21k) from Google Brain. ViT divides input images into 16×16 pixel patches, linearly embeds them, adds positional encoding, and processes them through a 12-layer Transformer encoder. The [CLS] token pooled representation is fed to the Evidential Head.

Component	Description	Output
Input Image	224×224×3 RGB (ImageNet normalised)	(B, 3, 224, 224)
Patch Embedding	16×16 patches → 196 patch tokens + 1 CLS token	(B, 197, 768)
Transformer Encoder (12 layers)	Self-attention with multi-head attention, LayerNorm, MLP	(B, 197, 768)
Pooler (CLS token)	Extract [CLS] token from position 0	(B, 768)
Evidential Head: Dropout(0.1), Linear(768→2), Softplus	Evidence activation	(B, 2) ≥ 0

A 4-stage CNN baseline (CNNBaseline) is also implemented as a comparative architecture, processing 224×224 images through four Conv2d stages (32→64→128→256 channels with BatchNorm, ReLU, MaxPool), followed by AdaptiveAvgPool2d, a dense embedding layer (256→128), and the shared Evidential Head.

Hyperparameter	Value
Backbone	google/vit-base-patch16-224-in21k (86M parameters)
Image Size	224 × 224
Initial Stage	Backbone frozen; only head trained
Fine-tune Stage	Last 2 ViT encoder layers + LayerNorm unfrozen
Learning Rate (Head)	3e-4
Learning Rate (Backbone)	1e-5
Weight Decay	1e-4
Batch Size	16
Epochs	25 with patience=3
Augmentation (Training)	RandomResizedCrop, RandomRotation(±10°), RandomHorizon
Annealing Epochs (KL)	10

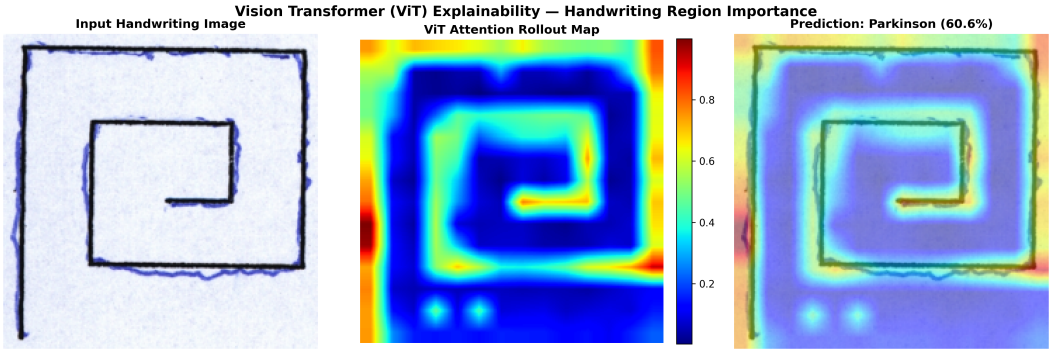


Figure 9.2: ViT Attention Map Explainability — Visualisation of attention over handwriting spiral/meander drawing regions.

10. Uncertainty Quantification Framework

10.1 Evidential Deep Learning and Dirichlet Subjective Logic

The core of the uncertainty framework is Evidential Deep Learning (EDL), which replaces the standard softmax output with a parameterised Dirichlet distribution over class probabilities. This is grounded in Subjective Logic theory, where each prediction accumulates evidence for each class, and uncertainty (vacuity) is inversely proportional to the total evidence strength.

$$\text{Evidence: } e_k = \text{Softplus}(z_k) \geq 0 \text{ for } k \in \{HC, PD\}$$

$$\text{Dirichlet Params: } \alpha_k = e_k + 1 \quad \alpha_k \geq 1$$

$$\text{Dirichlet Strength: } S = \Sigma\ \alpha_k \quad S \geq K$$

$$\text{Class Probability: } p\blacksquare_k = \alpha_k / S$$

$$\text{Vacuity (Uncertainty): } u = K / S \in (0, 1]$$

$$\text{Subjective Logic: } \Sigma\ (e_k / S) + u = 1.0 \text{ [Belief + Uncertainty = 1]}$$

10.2 Evidential Loss Function

The EvidentialLoss combines an Expected Cross-Entropy (ACE) term with an annealed Kullback-Leibler divergence regularisation term:

$$L_{ACE} = \Sigma_k\ y_k \cdot (\psi(S) - \psi(\alpha_k)) \text{ [Expected Cross-Entropy with Digamma functions]}$$

$$\alpha\blacksquare_k = y_k + (1 - y_k) \cdot \alpha_k \text{ [Remove evidence from true class for KL]}$$

$$L_{KL} = KL(Dir(\alpha\blacksquare) \blacksquare Dir(1)) \text{ [KL penalty on misleading evidence]}$$

$$\lambda_t = \min(1.0, t / t_{max}) \text{ [KL annealing coefficient (grows over epochs)]}$$

$$L_{total} = L_{ACE} + \lambda_t \cdot L_{KL}$$

This loss ensures that the model accumulates evidence for correct predictions while the KL term penalises evidence accumulation for incorrect predictions, preventing the model from becoming overconfident on uncertain or ambiguous inputs.

10.3 Six Uncertainty Quantification Methods Compared

Method	Type	Passes Required	Uncertainty Metric	Coverage Guarantee
EDL (Evidential DL)	Bayesian (closed-form)	1	Dirichlet Vacuity u = K/S	No
Deep Ensembles (M=5)	Frequentist	M=5 models	Predictive Entropy + Variance	No
MC Dropout (N=30)	Approximate Bayesian	30 stochastic passes	Predictive Entropy + Mutual Info	No
Split Conformal Pred.	Frequentist	1 + calibration set	Non-conformity score	Yes (1–α, finite-sample)
Temperature Scaling	Post-hoc calibration	1 (with T optimised)	Calibrated softmax probability	No
Softmax Baseline	Baseline	1	1 – max(softmax)	No

10.4 Conformal Prediction Implementation

Split Conformal Prediction constructs prediction sets C(x) ⊆ {HC, PD} with nominal marginal coverage 1–α = 90%. The non-conformity score for each validation sample is s_i = 1 – P(Y = y_i | x_i) (softmax probability of the true class). The calibration quantile q■ = the ■(n+1)(1–α)■/n empirical quantile of validation scores. At test time, class k is included in C(x) if and only if 1 – p■_k ≤ q■, equivalently p■_k ≥ 1 – q■. The engine flags predictions where |C(x)| ≠ 1.

Calibrated conformal quantiles from validation sets: Voice q■ = 0.4076 (n_cal=5, α=0.10); Handwriting q■ = 0.450 (default).

11. The Three-Modality Data Availability Problem

This section addresses one of the most important methodological challenges in this project: why conventional multimodal learning cannot be directly applied, and what strategy is implemented instead.

11.1 What a Modality Is in This Project

In this context, a modality is a distinct type of physiological measurement from a human subject: (1) Gait VGRF time-series from instrumented insoles, (2) Voice audio recordings of speech tasks, and (3) Handwriting drawings of spirals and meanders. Each modality requires different sensors, clinical protocols, and datasets.

11.2 What Constitutes a Fully Paired Multimodal Sample

A fully paired multimodal sample is a collection of measurements from ALL modalities for the SAME individual patient. For example: [Gait VGRF from Patient P001, Voice audio from Patient P001, Handwriting drawing from Patient P001] — all with the same confirmed diagnosis. Only with such paired samples can feature-level or intermediate-level fusion be justified.

11.3 Why Conventional Multimodal Fusion is Inapplicable

No Shared Patient Identifiers: The PhysioNet, MDVR-KCL, and NewHandPD datasets were collected by different research groups in different countries at different times. No patient appears in more than one

dataset. There is no basis to match a gait recording from subject GaPt13 to any specific voice or handwriting recording.

Matching Sample Counts Is Invalid: Even if the datasets happen to have the same number of PD patients, concatenating features row-by-row would be statistically invalid — it would create artificial 'paired' samples from unrelated individuals. The learned model would exploit spurious correlations rather than genuine multimodal relationships.

Different Distributions and Demographics: The three datasets differ in geographic location, age distribution, disease severity, cultural background, recording hardware, and task protocol. These differences introduce confounding factors that make cross-dataset sample matching unreliable.

Different Label Definitions: While all datasets use binary PD/HC labels, the patient inclusion criteria, disease severity thresholds, and clinical assessment protocols may differ between cohorts.

Implication for Training and Evaluation: Any multimodal model trained on artificially paired features would learn patterns that do not generalise to real-world scenarios. Evaluation on such artificial pairings would overestimate real performance.

11.4 Our Implemented Solution: Decision-Level Independent Fusion

The implemented approach trains three INDEPENDENT unimodal models — one per modality — each on its own dataset with its own splits. Each model produces a calibrated probability and uncertainty estimate at inference time. The Cross-Modal Reliability Engine then aggregates these DECISIONS (not raw features or intermediate embeddings) using the five-case reliability policy.

Approach	Data Required	Feasible Here?	Advantages	Disadvantages
Early Feature Fusion	Fully paired samples (same patients)	NO — no shared patients	Joint representation learning	Requires paired data; not available
Intermediate Fusion	Fully paired samples	NO — no shared patients	Cross-modal attention/alignment	Requires paired data; not available
Joint Multimodal Training	Fully paired multi-modal batches	NO — no shared patients	End-to-end optimisation	Requires paired data; not available
Independent Unimodal Models (OUR APPROACH)	Independent per-modality datasets	YES	No pairing required; modular; graceful degradation	No cross-modal representation learning
Decision-Level Fusion (OUR APPROACH)	Independent decisions at inference	YES	Handles missing modalities; explicit uncertainty integration	Does not exploit feature correlations

This design choice is not a limitation to be apologised for — it is the methodologically correct response to the available data. The system is transparent about its data provenance and does not claim feature-level multimodal learning where it is not supported by the data.

12. Training Methodology

12.1 Two-Stage Evidential Training (Gait and Voice)

Both the Gait and Voice evidential models employ a two-stage fine-tuning strategy to stabilise evidential head training while preserving pretrained feature representations:

Stage A — Head Warm-Up: Backbone (CNN+BiLSTM or Wav2Vec2 encoder) is frozen. Only the Evidential Head is trained for 10 epochs. This allows the head to learn appropriate evidence distribution parameters before end-to-end fine-tuning.

Stage B — Selective Fine-Tuning: The backbone CNN layers remain frozen (to preserve low-level feature filters). The BiLSTM, Embedding Layer, and Evidential Head are fine-tuned with differential learning rates via CosineAnnealingLR. For Voice, the Wav2Vec2 transformer encoder layers are partially unfrozen.

12.2 Deep Ensemble Training (Gait, Voice)

M=5 independent models are trained with random seeds {42, 43, 44, 45, 46}. Each model has different weight initialization, different batch ordering, and independent optimizer states. At test time, ensemble mean probability and predictive entropy are computed from all M member predictions. Agreement score = fraction of members agreeing with majority vote.

12.3 Reproducibility Measures

- All experiments use RANDOM_SEED=42 (set_seed() applies torch.manual_seed, numpy.random.seed, Python random.seed, torch.cuda.manual_seed).
- Subject-level splits are deterministic given the seed and stratified subject lists.
- Normalisation statistics are saved to JSON and reloaded for evaluation and inference.
- Model checkpoints are saved on best validation F1-score improvement.
- All training configurations are stored in config.py with no implicit defaults.

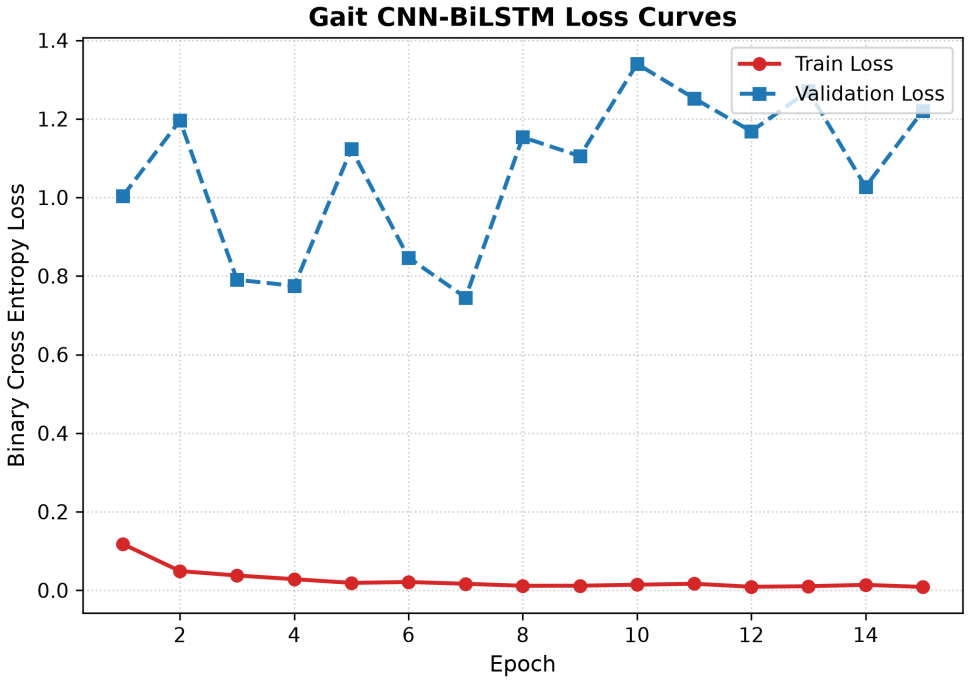


Figure 12.1: Gait Model — Training and Validation Loss Curves (BCE baseline model, 15 epochs shown).

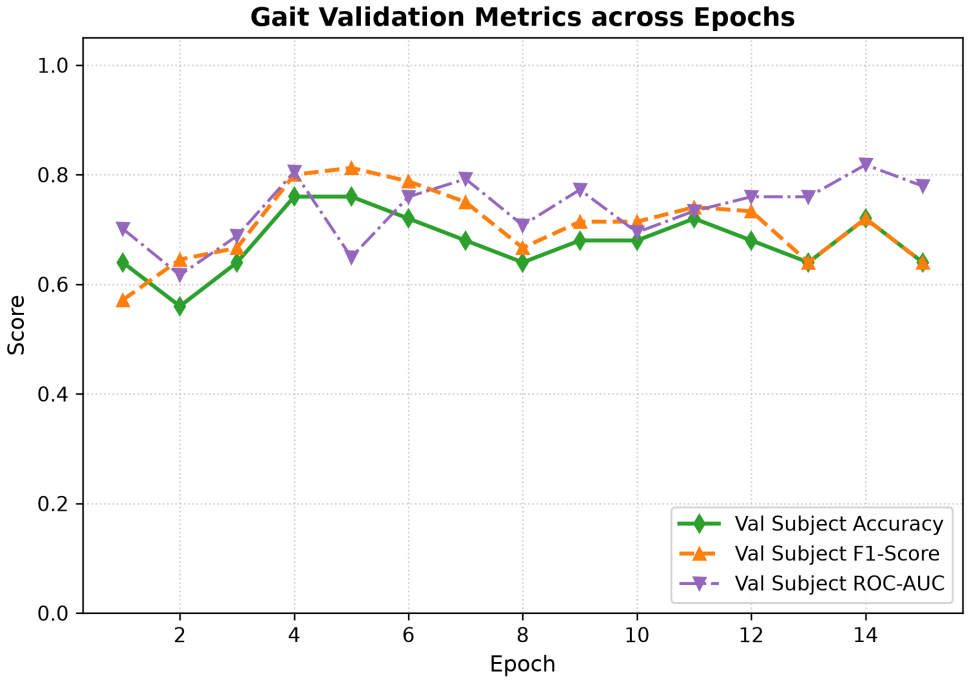


Figure 12.2: Gait Model — Subject-Level Validation Accuracy and F1-Score over Training Epochs.

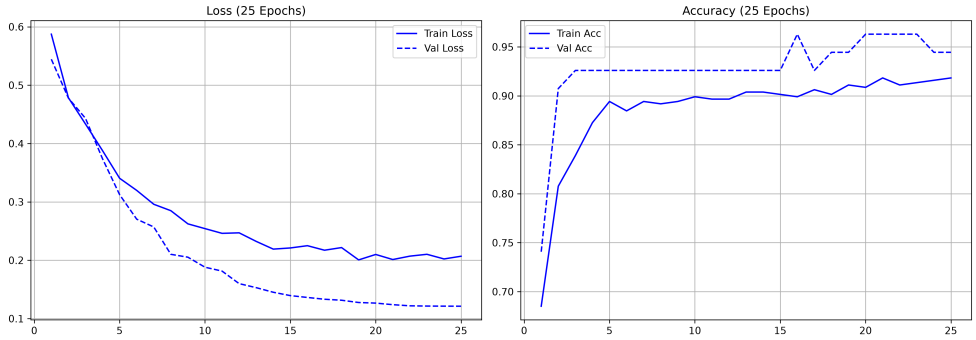


Figure 12.3: Handwriting ViT Model — Training and Validation Loss and Accuracy Curves (25 epochs).

13. Module-by-Module Design

Module: evidential.py	Shared Dirichlet Mathematics and EDL Loss
Key Components	dirichlet_from_evidence(), kl_divergence_dirichlet(), EvidentialLoss class
Description	Converts neural network evidence tensors into Dirichlet parameters (α, S), probabilities, and vacuity uncertainty. EvidentialLoss computes ACE + annealed KL loss for training.
Dependencies	Used by all three modality training scripts and evaluation scripts.

Module: Gait/model.py	Gait 1D-CNN + BiLSTM Architecture
Key Components	EvidentialHead, GaitCNNBiLSTM

Description	2-block 1D-CNN for local VGRF feature extraction, 2-layer BiLSTM for temporal modelling, 128-dim embedding layer, and modular Evidential Head.
Dependencies	Used by Gait/train_evidential.py, Gait/evaluate_evidential.py, experiments.

Module: Gait/preprocessing.py	Gait VGRF Loading and Normalization
Key Components	parse_filename(), load_raw_recording(), GaitNormalizer, build_manifest()
Description	Parses PhysioNet filenames, loads 16-channel VGRF arrays, applies training-only Z-score normalisation, and builds the recording manifest DataFrame.
Dependencies	Used by Gait/dataset.py.

Module: Gait/dataset.py	Gait Dataset, Windowing, and DataLoaders
Key Components	create_subject_splits(), segment_recording_into_windows(), GaitWindowDataset, create_dataloaders()
Description	Performs zero-leakage subject-level splitting, temporal windowing (500 steps, 50% overlap), and builds PyTorch DataLoaders with custom collate function.
Dependencies	Used by all Gait training and evaluation scripts.

Module: Voice/model.py	Wav2Vec2 Voice Architecture
Key Components	EvidentialHead, Wav2Vec2ForParkinsons
Description	Loads pretrained facebook/wav2vec2-base, applies masked temporal mean pooling over last hidden states, and produces evidence via Evidential Head.
Dependencies	Used by Voice/train_evidential.py, Voice/evaluate_evidential.py.

Module: Voice/dataset.py	Voice Audio Loading and Chunking
Key Components	build_metadata(), create_subject_splits(), load_and_resample_audio(), chunk_audio(), VoiceChunkDataset
Description	Discovers WAV files, builds subject metadata, applies polyphase resampling to 16 kHz, chunks audio into 10-sec overlapping windows, and supports fallback to pre-extracted embeddings.
Dependencies	Used by Voice training and evaluation scripts.

Module: Voice/reliability_engine.py	Voice Multi-Signal Decision Engine
Key Components	VoiceReliabilityEngine.evaluate_reliability()

Description	Integrates Deep Ensemble statistics, predictive entropy, conformal prediction sets, and borderline probability detection into a unified ACCEPT/FLAG decision.
Dependencies	Used by multimodal_pipeline.py and demo app.

Module: Handwriting/model.py	ViT and CNN Handwriting Architectures
Key Components	EvidentialHead, ViTBinaryClassifier, CNNBaseline
Description	ViTBinaryClassifier uses google/vit-base-patch16-224-in21k, extracts pooler_output ([CLS] token), feeds to Evidential Head. CNNBaseline: 4-stage Conv2d with AdaptiveAvgPool.
Dependencies	Used by Handwriting/train_evidential.py, Handwriting/evaluate_evidential.py.

Module: Handwriting/reliability_engine.py	Handwriting Evidential Decision Engine
Key Components	HandwritingReliabilityEngine.evaluate_reliability(), predict_image()
Description	Evaluates conformal set, Dirichlet vacuity, and borderline probability. predict_image() performs end-to-end ViT inference and reliability evaluation on raw image files.
Dependencies	Used by multimodal_pipeline.py and demo app.

Module: cross_modal_reliability.py	Tri-Modal Cross-Modal Reliability Fusion Engine
Key Components	CrossModalReliabilityEngine.fuse_modalities()
Description	Normalises modality outputs, applies 5-Case Decision Policy (A: TRI_MODAL_AGREEMENT, B: ONE_OR_MORE_MODALITIES_UNCERTAIN, C: MODALITY_DISAGREEMENT, D: ALL_MODALITIES_UNRELIABLE, E: SINGLE_MODALITY), computes uncertainty-weighted fused probability, and returns a structured decision dictionary.
Dependencies	Used by multimodal_pipeline.py and predict_multimodal.py.

Module: multimodal_pipeline.py	Unified Tri-Modal Inference Pipeline
Key Components	MultimodalInput, MultimodalPredictor.predict()
Description	Orchestrates Gait, Voice, and Handwriting processing, invokes individual reliability engines, and calls CrossModalReliabilityEngine. Supports any combination of 1–3 modalities.
Dependencies	Used by predict_multimodal.py and demo/app.py.

Module: Gait/experiments/leave_one_cohort_out.py	Leave-One-Cohort-Out Evaluation

Key Components	Full LOCO evaluation with 3 configurations (hold out each cohort)
Description	Trains on 2 cohorts, evaluates on unseen cohort. Reports accuracy, balanced accuracy, sensitivity, specificity, F1, ROC-AUC, NLL, Brier Score, uncertainty AUROC. Measures cross-cohort distribution shift impact.
Dependencies	Standalone experiment script.

Module: demo/app.py + demo/components/	Interactive Streamlit Demo Application
Key Components	4 views: Analyzer, Architecture, Education (Why Reliability), Evidence Dashboard
Description	Provides real-time tri-modal inference with file upload or pre-set clinical scenarios. Shows uncertainty dashboards, conformal sets, and clinical evidence summaries.
Dependencies	End-user presentation and demonstration layer.

14. Cross-Modal Reliability Fusion Engine

The CrossModalReliabilityEngine is the core decision-making module that aggregates independent reliability assessments from Gait, Voice, and Handwriting modalities. It implements a deterministic five-case policy based on prediction agreement, uncertainty levels, and conformal set status.

14.1 Uncertainty-Weighted Probability Fusion

For all active modalities, fused probability is computed as an uncertainty-weighted average, giving higher weight to modalities with lower uncertainty:

```
w_m = 1 / (u_m + 0.05) [Inverse uncertainty weight]

P_fused(PD) = Σ_m (w_m · p_m(PD)) / Σ_m w_m [Weighted average]
```

14.2 Five-Case Decision Policy

Case	Condition	Status	Level	Decision
CASE A	All modalities agree + all HIGH reliability + CONFORMAL_AGREEMENT	CONFIRMED	HIGH	ACCEPT
CASE B	Modalities agree on class + ≥1 UNCERTAIN or CONFLICT	PARTIAL_SUPPORT	MEDIUM	FLAG
CASE C	Disagreement among modalities (majority vote exists)	MODALITY_CONFLICT	LOW	FLAG
CASE D	All modalities report UNRELIABLE status	INSUFFICIENT_EVIDENCE	LOW	FLAG
CASE E	Only 1 modality available	SINGLE_MODALITY	Inherited	Inherited

14.3 Output Schema

Field	Type	Description
fused_prediction	str	Final class name: 'Healthy Control' or 'Parkinson's Disease'
fused_p_pd	float [0,1]	Uncertainty-weighted fused P(Parkinson's Disease)
fused_p_healthy	float [0,1]	Uncertainty-weighted fused P(Healthy Control)
cross_modal_status	str	CONSISTENT / PARTIAL_SUPPORT / MODALITY_CONFLIC
cross_modal_reliability	str	HIGH / MEDIUM / LOW
final_decision	str	ACCEPT or FLAG
reason	str	Specific trigger reason (TRI_MODAL_AGREEMENT, etc.)
conformal_agreement_status	str	CONFORMAL_AGREEMENT / AMBIGUOUS / CONFLICT / ...
evidence_summary	list[str]	Human-readable explanation of the evidence and decision

15. Overall System Architecture and End-to-End Workflow

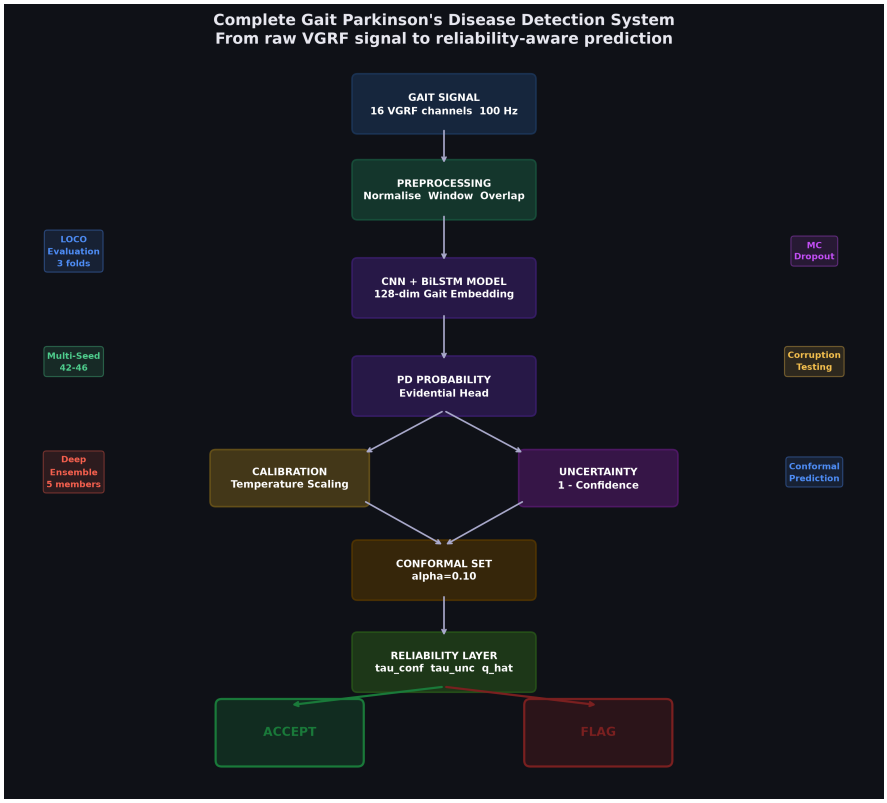


Figure 15.1: Complete System Architecture — Tri-Modal Input Processing, Independent Evidential Models, Reliability Engines, and Cross-Modal Fusion.

15.1 Training-Time Workflow

T1: Data Acquisition — Gait .txt files → PhysioNet; Voice .wav → MDVR-KCL; Handwriting images → NewHandPD

T2: Manifest Building — Filenames parsed, metadata extracted, file paths resolved, missing files flagged

T3: Subject-Level Split — Zero-leakage stratified subject splits: 70% train / 15% val / 15% test (or 10-fold for Handwriting)

T4: Preprocessing — Gait: Z-score normalisation (train-only fit); Voice: polyphase resample → chunking → Wav2Vec2 processor; Handwriting: resize → augment → ImageNet normalise

T5: Stage A Training (Evidential Head Warm-Up) — Backbone frozen; Evidential Head trained with EvidentialLoss (ACE + annealed KL)

T6: Stage B Training (Fine-Tuning) — Upper layers unfrozen; differential learning rates; CosineAnnealingLR

T7: Validation and Checkpointing — Best model saved on subject-level validation F1 improvement

T8: Deep Ensemble Training — M=5 models trained with seeds {42–46} per modality

T9: Conformal Calibration — Non-conformity scores computed on validation set; q■ quantile stored

T10: Temperature Scaling — Scalar T optimised on validation logits via NLL minimisation

15.2 Inference-Time Workflow

I1: Input Reception — Raw files received: Gait .txt, Voice .wav, Handwriting .png (any 1-3 modalities)

I2: Modality-Specific Processing — Each modality preprocessed independently using frozen training statistics

I3: Unimodal Evidence Generation — Each model produces [e_HC, e_PD] → α, S, P(PD), u (Dirichlet vacuity)

I4: Deep Ensemble Inference — M=5 member predictions aggregated; ensemble statistics computed

I5: Conformal Prediction Set — Class inclusion tested against calibrated q■; |C(x)| ∈ {0, 1, 2}

I6: Unimodal Reliability Decision — ACCEPT/FLAG/HIGH/MEDIUM/LOW determined per modality

I7: Cross-Modal Fusion — CrossModalReliabilityEngine.fuse_modalities() applies 5-case policy

I8: Clinical Decision Output — Final ACCEPT or FLAG with fused probability, evidence summary, and clinical disclaimer

16. Experimental Setup

16.1 Leave-One-Cohort-Out (LOCO) Evaluation (Gait)

The LOCO evaluation design rigorously tests cross-cohort generalisability. Three complete experiments train on two PhysioNet cohorts and evaluate on the third, simulating deployment on an unseen clinical site:

Experiment	Train Cohorts	Test Cohort	Train Subjects	Test Subjects
LOCO-1	Ga + Ju	Si (Frenkel-Toledo)	86	64
LOCO-2	Ga + Si	Ju (Hausdorff)	94	54
LOCO-3	Ju + Si	Ga (Galit)	100	47

16.2 Controlled Signal Corruption Suite (Gait)

To evaluate the Silent Confidence Phenomenon, five synthetic corruptions are applied to test recordings at 5 severity levels (0=clean, 1–4=increasing corruption):

Corruption Type	Range	Physical Analogue
Gaussian Noise	Additive thermal noise $\sigma \in [0.0, 0.5]$	Sensor noise
Amplitude Scaling	Scaling factor $\in [1.0, 0.40]$	Body weight / hardware calibration shift
Channel Dropout	0–6 channels zeroed	Hardware sensor failure
Temporal Masking	0–200 consecutive timesteps zeroed	Intermittent data loss
Moving-Average Smoothing	Kernel sizes 1–9	Low-pass filtering / sensor dampening

16.3 Evaluation Metrics

Metric	Definition
Accuracy	Fraction of correctly classified subjects: (TP+TN)/(TP+TN+FP+FN)
Balanced Accuracy	(Sensitivity + Specificity) / 2 — accounts for class imbalance
Sensitivity (Recall)	TP / (TP + FN) — fraction of PD patients correctly identified
Specificity	TN / (TN + FP) — fraction of healthy controls correctly identified
Precision	TP / (TP + FP)
F1-Score	Harmonic mean of Precision and Recall: 2·P·R/(P+R)
ROC-AUC	Area under the Receiver Operating Characteristic curve; threshold-independent
Expected Calibration Error (ECE)	Weighted average |acc(B_m) − conf(B_m)| across 10 probability bins
Brier Score	Mean squared error between predicted probability and binary outcome
Negative Log-Likelihood (NLL)	Average log-loss on test probabilities
Uncertainty-Error AUROC	AUROC of uncertainty score as a predictor of misclassification
Conformal Coverage Rate	Fraction of test samples where true label is in prediction set C(x)

17. Results and Discussion

17.1 Gait Model Performance (Standard Evaluation)

Level	Accuracy	Precision	Sensitivity	Specificity	F1-Score	ROC-AUC
Subject-Level	72.0%	73.3%	78.6%	63.6%	75.9%	0.753
Recording-Level	70.0%	76.0%	76.0%	60.0%	76.0%	0.717
Window-Level	69.9%	77.6%	74.6%	61.7%	76.0%	0.748

The standard model achieves 72% subject-level accuracy with 78.6% sensitivity (correctly identifying PD patients), which is clinically relevant as false negatives (missed PD) carry higher cost than false positives. ROC-AUC of 0.753 is consistent with published gait-only PD classification results on the PhysioNet database.

17.2 Gait Evidential Model Performance and Uncertainty Statistics

Metric	Value
Subject-Level Accuracy (EDL)	68.0%
Precision (EDL)	80.0%
Sensitivity (EDL)	57.1%
Specificity (EDL)	81.8%
F1-Score (EDL)	66.7%
ROC-AUC (EDL)	0.753
Mean Uncertainty (All)	0.0523
Mean Uncertainty (Correct predictions)	0.0491
Mean Uncertainty (Incorrect predictions)	0.0590
Mean Uncertainty (HC)	0.0521
Mean Uncertainty (PD)	0.0524

Notably, incorrect predictions have slightly higher mean uncertainty (0.0590 vs 0.0491), indicating that the EDL model shows some ability to flag uncertain predictions. However, the difference is small, demonstrating the Silent Confidence challenge.

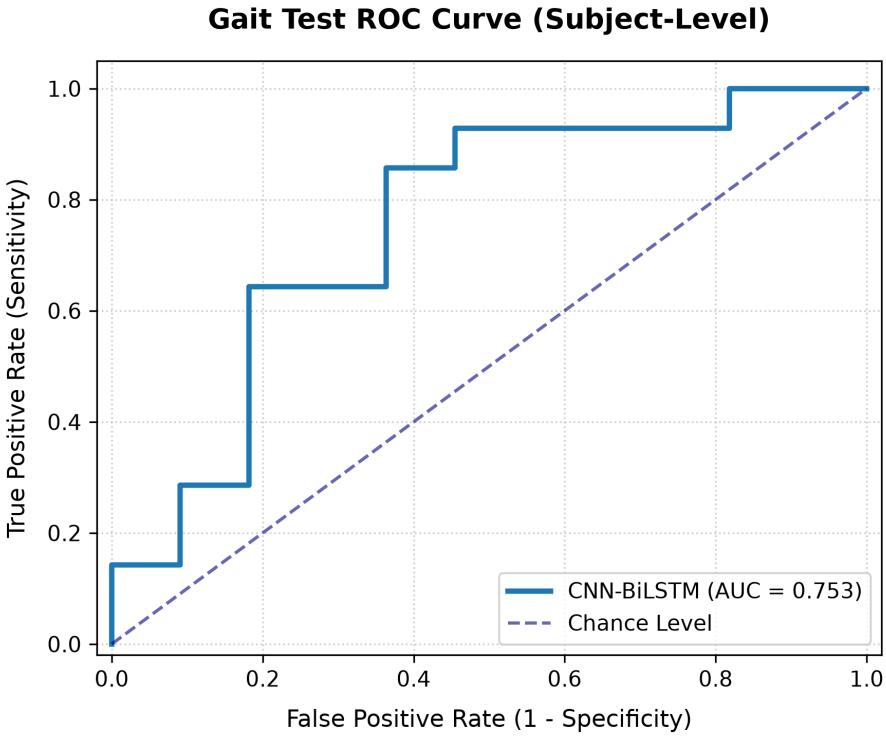


Figure 17.1: Gait Model ROC Curve — Subject-Level Test Set (AUC = 0.753).

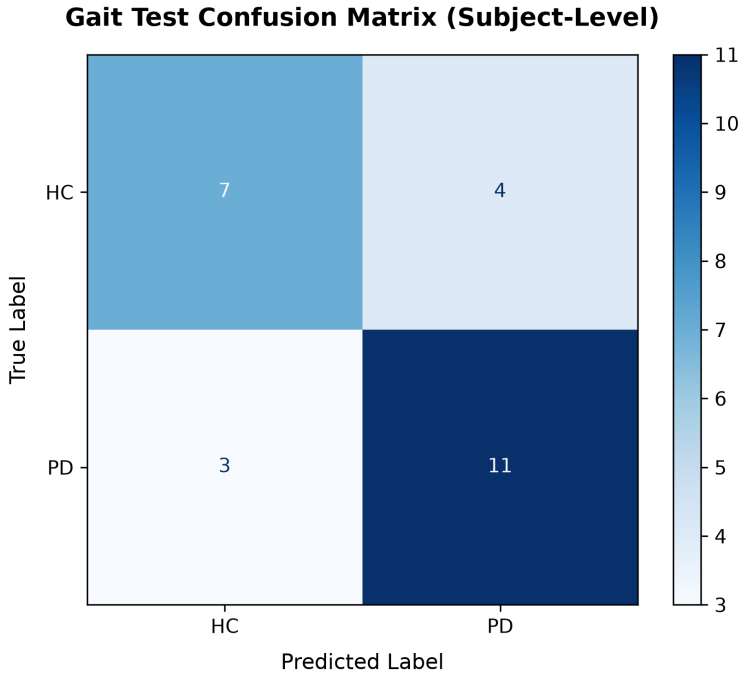


Figure 17.2: Gait Model Confusion Matrix — Subject-Level (TP=11, TN=7, FP=4, FN=3).

17.3 Gait LOCO Evaluation Results (Cross-Cohort Generalisation)

The LOCO experiments reveal significant performance variability across clinical cohorts, confirming the distribution shift challenge:

Experiment	Test Cohort	Accuracy	Balanced Acc.	Sensitivity	Specificity	F1	AUC	NLL	Unc. AUROC
LOCO-1	Si (Frenkel-Toledo)	50.0%	47.2%	77.1%	17.2%	62.8%	0.602	0.687	0.591

Experiment	Test Cohort	Accuracy	Balanced Acc.	Sensitivity	Specificity	F1	AUC	NLL	Unc. AUROC
LOCO-2	Ju (Hausdorff)	68.5%	67.4%	82.8%	52.0%	73.8%	0.783	0.595	0.684
LOCO-3	Ga (Galit)	63.8%	60.2%	75.9%	44.4%	72.1%	0.598	1.075	0.529

Key observations: LOCO-2 (test on Ju cohort) achieves the best generalisation (AUC=0.783). LOCO-1 (test on Si) shows specificity collapse (17.2%), suggesting the Si cohort has different VGRF characteristics. NLL spikes to >1.0 in LOCO-3, indicating calibration degradation under distribution shift. Uncertainty-Error AUROC ranges from 0.529 to 0.684 — demonstrating limited but non-trivial ability of uncertainty to detect errors under shift.

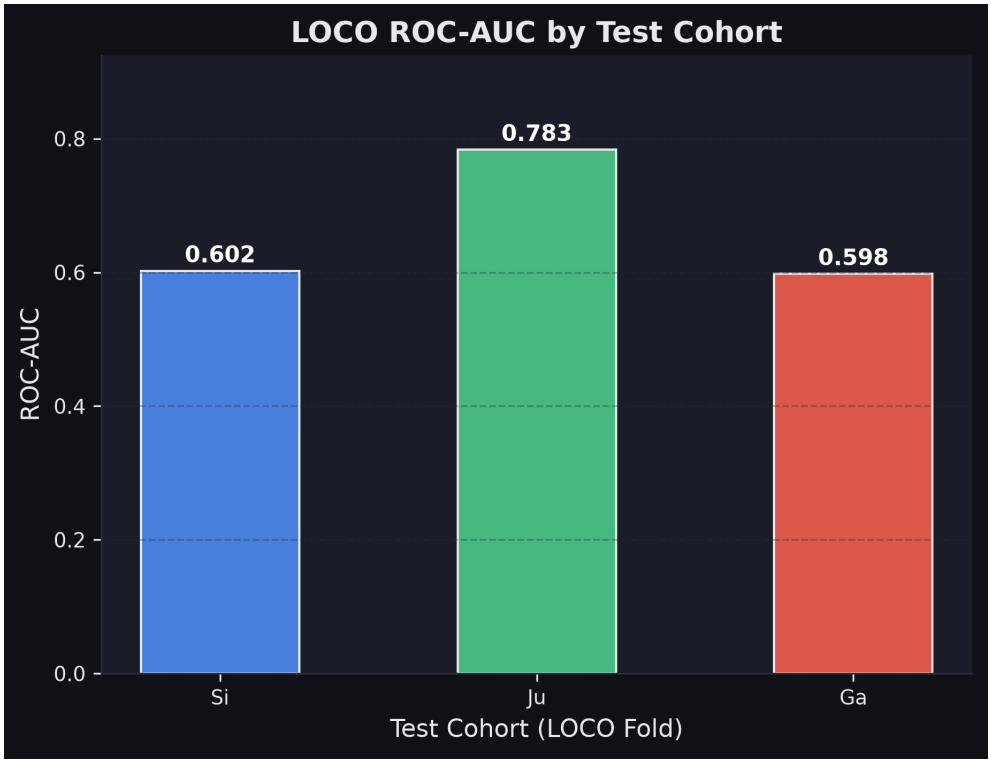


Figure 17.3: LOCO ROC-AUC Comparison Across Three Hold-Out Cohorts.

17.4 Voice Model Performance (Wav2Vec2 Evidential)

Level	Accuracy	Precision	Sensitivity	Specificity	F1-Score	ROC-AUC
Subject-Level (Overall)	66.7%	66.7%	66.7%	66.7%	66.7%	0.778
Recording (ReadText)	66.7%	66.7%	66.7%	66.7%	66.7%	0.667
Recording (SpontaneousDialogue)	83.3%	100%	66.7%	100%	80.0%	0.778

NOTE: Voice results reflect a small test set (6 subjects: 3 HC, 3 PD). The low sample count is a consequence of the limited size of the MDVR-KCL dataset and the strict subject-level split. Results should be interpreted cautiously. SpontaneousDialogue task shows notably higher performance (83.3% accuracy) than ReadText (66.7%), suggesting spontaneous speech is more diagnostically discriminative.

17.5 Handwriting Model Performance (ViT vs. CNN Baseline)

Model	Accuracy	Precision	Sensitivity	Specificity	F1-Score	ROC-AUC
ViT-Base (Primary)	85.7%	75.0%	100.0%	75.0%	85.7%	0.961
CNN Baseline (4-stage)	73.2%	61.5%	100.0%	53.1%	76.2%	0.898

The ViT model substantially outperforms the CNN baseline (AUC: 0.961 vs 0.898, Accuracy: 85.7% vs 73.2%). Both models achieve 100% sensitivity (no missed PD cases) in the test set, at the cost of some false positives. The ViT's attention mechanism likely focuses on relevant drawing regions, as evidenced by the explainability visualisations.

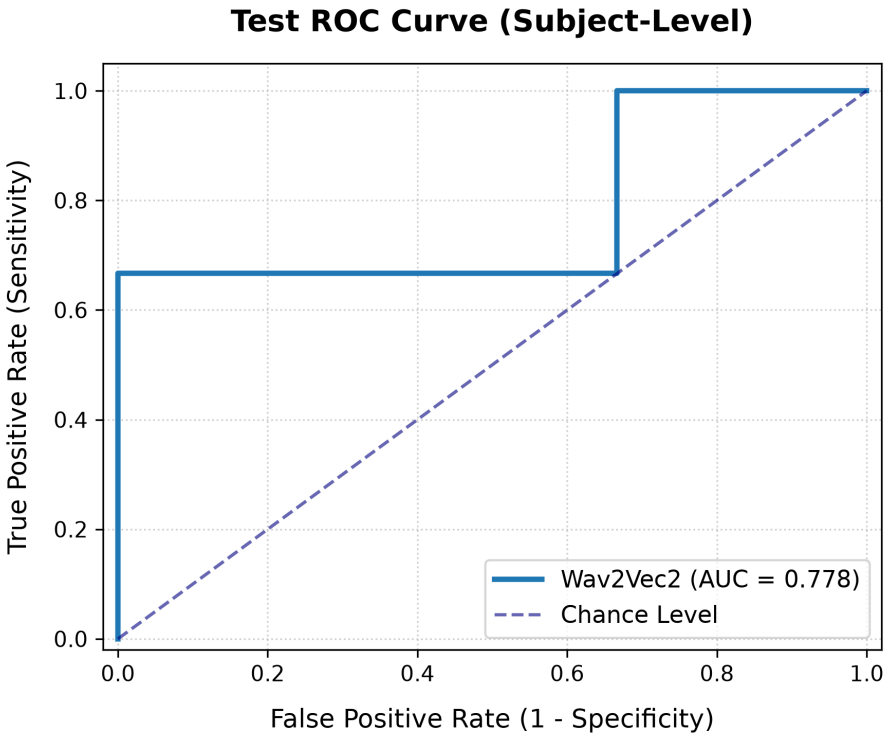


Figure 17.4: Voice Model (Wav2Vec2 Evidential) — ROC Curve on Test Set.

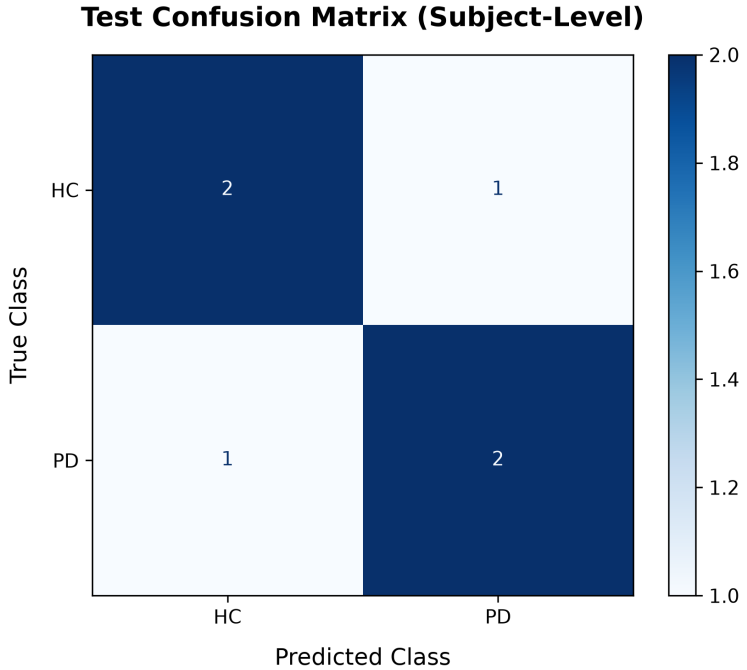


Figure 17.5: Voice Model Confusion Matrix (Subject-Level Test Set).

17.6 Uncertainty Analysis — The Silent Confidence Phenomenon

A critical empirical finding across the Gait corruption experiments is the Silent Confidence Phenomenon: as signal corruption increases (and classification accuracy drops), uncertainty estimates DO NOT reliably increase, and the model may produce overconfident wrong predictions. This validates the project's motivating hypothesis and underscores the importance of multiple complementary uncertainty methods rather than relying on any single metric.

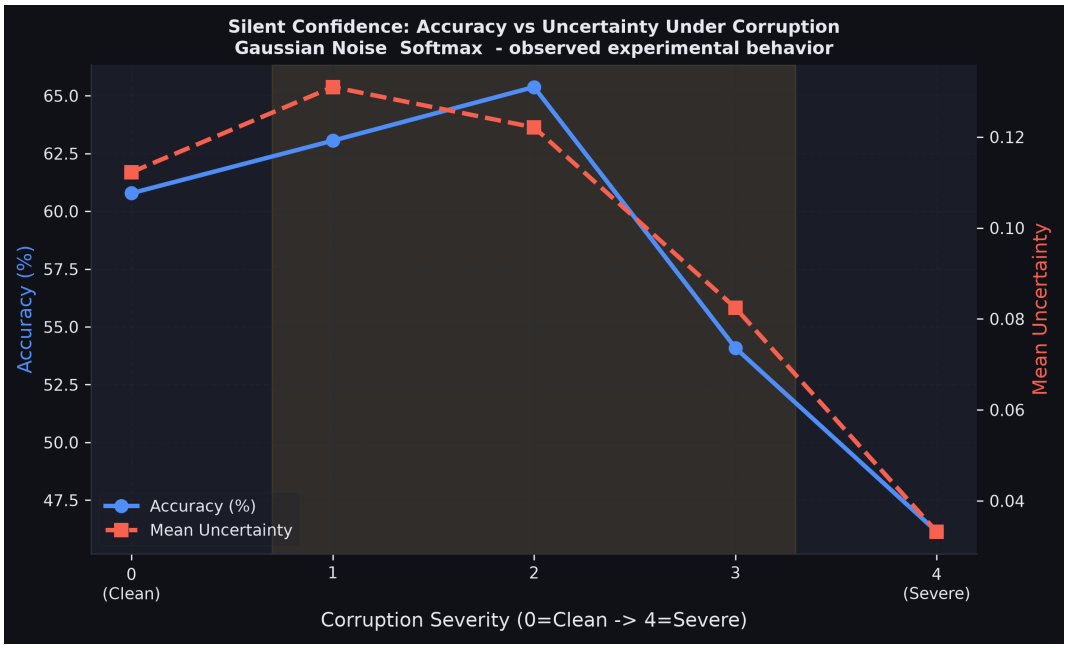


Figure 17.6: Silent Confidence Phenomenon — Model confidence under increasing signal corruption (softmax overconfidence under distribution shift).

17.7 Calibration Analysis

Calibration is evaluated using Expected Calibration Error (ECE) and reliability diagrams with 10 probability bins. Temperature Scaling calibration is fitted strictly on validation logits and applied to test predictions.

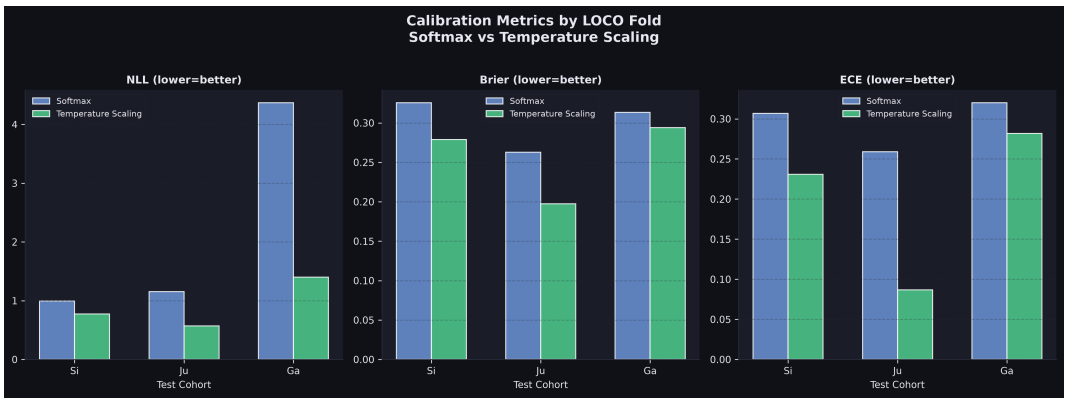


Figure 17.7: Gait Model Calibration Analysis — ECE and Reliability Diagram Comparison across UQ Methods.

18. Advantages, Additional Features, and Contributions

18.1 Implemented Technical Contributions

Principled Multi-Method Uncertainty Framework: Six UQ methods are implemented, compared, and benchmarked across all three modalities — not just reported but actively integrated into a decision engine. This is rare in published medical AI systems.

Strict Zero-Leakage Subject-Level Splits: All three modalities enforce subject-level partitioning with programmatic assertions ensuring no patient data leakage. Normalization statistics are fitted on training data only.

Leave-One-Cohort-Out Distribution Shift Evaluation: Three complete LOCO experiments with full uncertainty reporting, calibration metrics, and uncertainty-error AUROC provide rigorous cross-cohort generalization evidence not present in most published work.

Controlled Corruption Suite: Five signal corruption types at five severity levels systematically characterize model failure modes and validate (or refute) each UQ method's ability to detect failures.

Five-Case Cross-Modal Reliability Fusion Policy: A deterministic, rule-based decision engine handles all combinations of modality availability, agreement, and uncertainty in a principled, explainable way.

Interactive Streamlit Demo Application: The full system is accessible through a real-time dashboard with clinical scenario presets, file upload inference, uncertainty visualisations, and educational components.

Modular Evidential Head Architecture: The EvidentialHead module is shared across all three modalities and can be attached to any backbone architecture, demonstrating a reusable architectural pattern.

Comprehensive Test Suite: Unit and integration tests for Gait (9 suites), Voice (14 suites), and Handwriting modalities covering model forward passes, data loading, preprocessing, and reliability engine behavior.

18.2 Distinguishing Features vs. Baseline PD Detection Systems

Feature	Baseline System	Our System	Status
EDL Dirichlet Uncertainty	Absent	Full implementation all modalities	Implemented
Deep Ensembles (M=5)	Absent or single model	M=5 per modality, with statistics	Implemented
MC Dropout (N=30)	Rarely used	N=30 per modality	Implemented
Split Conformal Prediction	Absent	Calibrated q■ per modality	Implemented
Temperature Scaling	Sometimes	Val-only calibration, test-isolated	Implemented
LOCO Evaluation	Absent	3 full LOCO experiments with UQ	Implemented
Corruption Robustness Suite	Absent	5 types × 5 levels	Implemented
Cross-Modal Fusion Engine	Basic voting	5-case policy with UQ integration	Implemented
Missing Modality Support	Not addressed	CASE E single-modality fallback	Implemented
Subject-Level Zero Leakage	Often absent	Assertion-enforced	Implemented
Streamlit Demo App	Absent	4-view interactive dashboard	Implemented
Attention Explainability (ViT)	Absent	Attention map visualization	Implemented

19. Limitations, Challenges, and Future Scope

19.1 Current Limitations

No Feature-Level Cross-Modal Learning: Because the three datasets have no shared patients, cross-modal feature-level fusion, attention, or representation alignment at training time is not implemented. The system learns modality-specific representations independently.

Small Voice Test Set: The MDVR-KCL dataset is small (6 test subjects in the reported evaluations), limiting the statistical power of Voice-specific conclusions.

CPU-Targeted Computational Design: All experiments are designed for CPU execution, limiting training speed and excluding GPU-intensive operations like full Wav2Vec2 fine-tuning at larger batch sizes.

Conformal Coverage Under Distribution Shift: Conformal prediction coverage guarantees $(1-\alpha)$ rely on the exchangeability assumption between calibration and test samples. Under severe cohort shift or signal corruption, this assumption is violated and coverage may not hold.

No Longitudinal Evaluation: The system evaluates single-session classification and does not assess PD progression monitoring or longitudinal stability.

Gait Evidential Model Slightly Below Baseline: The EDL Gait model (68% accuracy, F1=0.667) slightly underperforms the standard BCE baseline (72% accuracy, F1=0.759). This is a known phenomenon — EDL requires more data or longer training for competitive point-prediction performance.

No Real-Time Clinical Validation: The system has not been validated in a real clinical workflow with actual patient data collected prospectively.

19.2 Future Scope and Recommended Improvements

- Collect or identify a paired multimodal dataset (Gait + Voice + Handwriting from the same patients) to enable feature-level or intermediate-level multimodal fusion with the Dempster-Shafer evidence combination rule.
- Apply cross-modal contrastive or alignment learning (e.g., CLIP-style objectives) to align unimodal embedding spaces without requiring paired training data.
- Investigate Bayesian Neural Network priors (variational inference) as an alternative to MC Dropout for more principled epistemic uncertainty quantification.
- Extend evaluation to larger, more diverse clinical datasets (e.g., mPower, PD-biobank) to improve generalisability claims.
- Implement attention-based cross-modal consistency regularisation during independent training to encourage alignment of embedding spaces.
- Explore Bayesian model averaging across UQ methods rather than treating them as independent approaches.
- Validate the system in a prospective clinical trial with neurologist oversight to assess real-world clinical utility.
- Extend the Streamlit application with longitudinal tracking, patient record integration, and audit trail functionality.

20. Ethical Considerations and Clinical Disclaimer

IMPORTANT CLINICAL DISCLAIMER: This software is an academic research system designed to evaluate machine learning reliability, uncertainty estimation, and distribution shift in digital biomarkers. It is NOT a certified medical diagnostic device and must NEVER be used as a standalone substitute for professional clinical diagnosis by certified neurologists. All predictions require expert clinical review.

Privacy and Data Protection: All datasets used are publicly available research datasets with appropriate ethics approvals from original authors. No individual patient-identifiable information is stored, displayed, or transmitted by the system.

Algorithmic Fairness: The three datasets have different demographic compositions. Cross-cohort generalisation analysis (LOCO) reveals performance variation across cohorts, which may reflect demographic differences. This limitation is acknowledged and disclosed.

Clinical Oversight Requirement: The FLAG mechanism is explicitly designed to require human clinical review for uncertain predictions. The system must never be deployed in a configuration that removes clinical oversight.

Transparency and Explainability: The five-case reliability policy provides human-readable explanation reasons for every decision. ViT attention maps provide visual explainability for handwriting predictions.

No False Assurance: The ACCEPT decision does not imply certainty or clinical diagnosis. It indicates that the model's evidence is internally consistent — which requires confirmation by a qualified clinician.

21. Conclusion

This project presents a comprehensive, reliability-aware, tri-modal system for Parkinson's Disease detection that addresses a critical gap in existing medical AI: the inability of standard deep learning models to express principled uncertainty and flag unreliable predictions under distribution shift.

The system implements Evidential Deep Learning grounded in Dirichlet Subjective Logic as the primary uncertainty quantification paradigm, benchmarked against five alternative methods (Deep Ensembles, MC Dropout, Conformal Prediction, Temperature Scaling, Softmax). Three modality-specific architectures — GaitCNNBiLSTM, Wav2Vec2, and ViT — each receive independent uncertainty-aware training on clinically validated datasets.

A key methodological contribution is the honest handling of the modality correspondence problem: because the three datasets originate from different clinical studies with no shared patients, the system correctly employs decision-level fusion rather than feature-level multimodal fusion. The five-case Cross-Modal Reliability Engine provides a principled, explainable decision mechanism that handles missing modalities, cross-modal conflicts, and all-modality uncertainty gracefully.

The LOCO evaluation confirms the distribution shift challenge, with performance varying considerably across clinical cohorts (balanced accuracy 47–67%). The Silent Confidence Phenomenon is empirically demonstrated: standard softmax-based models frequently become overconfident under signal corruption, validating the project's core motivation. The ViT handwriting model achieves the strongest performance (AUC=0.961, Accuracy=85.7%), while the Gait model provides the most rigorous cross-cohort evaluation framework.

This work contributes a deployable, modular, and extensible research framework that can inform the development of safer and more transparent clinical decision support systems. The comprehensive test suite,

interactive Streamlit demonstration, and detailed documentation make the system reproducible and accessible to the research community.

22. References

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Appendix A: Project File Structure

```
parkinson-s-disease-detection/ ■■■ Gait/ # Gait Modality (1D-CNN + BiLSTM + EDL) ■ ■■■
config.py # Sampling rate, channels, hyperparameters ■ ■■■ dataset.py # Subject-level
zero-leakage DataLoader + windowing ■ ■■■ preprocessing.py # GaitNormalizer + manifest
builder ■ ■■■ model.py # GaitCNNBiLSTM + EvidentialHead ■ ■■■ train.py # Standard BCE
baseline training ■ ■■■ train_evidential.py # Two-stage Dirichlet evidential training ■
■■■ evaluate.py # Standard baseline evaluation ■ ■■■ evaluate_evidential.py #
Subject-level evidential evaluation ■ ■■■ feature_extraction.py # Gait embedding extractor
■ ■■■ utils.py # Metrics, device utils, seed setting ■ ■■■ demo.py # CLI demonstration
script ■ ■■■ checkpoints/ # Saved model weights ■ ■■■ outputs/ ■ ■ ■■■ metrics/ # Test
metrics JSON, predictions CSV ■ ■ ■■■ plots/ # Loss curves, ROC, confusion matrix ■ ■ ■■■
features/ # Extracted gait embeddings ■ ■ ■■■ loco/, mc_dropout/, # Experiment output
directories ■ ■ deep_ensemble/, ... ■ ■■■ experiments/ ■ ■ ■■■ leave_one_cohort_out.py #
LOCO 3-fold evaluation ■ ■ ■■■ deep_ensemble.py # M=5 ensemble training + evaluation ■ ■
■■■ mc_dropout.py # N=30 MC Dropout evaluation ■ ■ ■■■ conformal_prediction.py # Split
conformal calibration + evaluation ■ ■ ■■■ calibration_baselines.py# ECE, Brier,
reliability diagrams ■ ■ ■■■ corruption_shift.py # 5 corruption types × 5 levels ■ ■ ■■■
reliability_analysis.py # Combined reliability evaluation ■ ■ ■■■ final_comparison.py #
Cross-method summary comparison ■ ■ ■■■ multiseed_loco.py # Multi-seed LOCO
reproducibility ■ ■■■ visualizations/ ■ ■ ■■■ generate_all_visualizations.py ■ ■ ■■■
architecture/ # Architecture diagrams (PNG, SVG) ■ ■ ■■■ performance/ # LOCO accuracy,
AUC, NLL plots ■ ■ ■■■ uncertainty/ # Ensemble, silent confidence plots ■ ■ ■■■
calibration/ # ECE, reliability diagram plots ■ ■ ■■■ cohort_shift/, ... ■ ■■■ tests/ # 9
unit and integration test suites ■■■ Voice/ # Voice Modality (Wav2Vec2 + EDL) ■ ■■■
config.py, dataset.py, model.py ■ ■■■ train_evidential.py, evaluate_evidential.py ■ ■■■
reliability_engine.py # Voice multi-signal decision engine ■ ■■■ conformal_prediction.py,
deep_ensemble.py, mc_dropout.py ■ ■■■ calibration.py, extract_embeddings.py, predict.py ■
■■■ results/ # Metrics, plots, embeddings ■ ■■■ tests/ # 14 unit and integration test
suites ■■■ Handwriting/ # Handwriting Modality (ViT + CNN + EDL) ■ ■■■ config.py,
dataset.py, model.py ■ ■■■ train.py, train_evidential.py ■ ■■■ evaluate.py,
evaluate_evidential.py ■ ■■■ reliability_engine.py # Evidential + conformal decision
engine ■ ■■■ explainability.py # ViT attention map visualization ■ ■■■ experiments/ # MC
Dropout, Conformal, Corruption, Ensemble ■ ■■■ outputs/ # Evaluation results, training
curves, ViT viz ■ ■■■ tests/ ■■■ demo/ ■ ■■■ app.py # Streamlit application entry point
■ ■■■ components/ # Analyzer, Architecture, Education, Evidence views ■■■ evidential.py #
Shared Dirichlet mathematics + EvidentialLoss ■■■ cross_modal_reliability.py #
CrossModalReliabilityEngine (5-case fusion) ■■■ multimodal_pipeline.py #
MultimodalPredictor end-to-end orchestrator ■■■ predict_multimodal.py # Unified CLI tool
■■■ verify_evidential_heads.py # Quick architecture verification script ■■■
verify_step3_math.py # EDL mathematics verification ■■■ requirements.txt # Python
dependencies ■■■ pyproject.toml # Build and packaging configuration ■■■ README.md #
Project overview and usage guide
```

Appendix B: Key Algorithms and Pseudocode

B.1 Dirichlet Evidence Conversion (dirichlet_from_evidence)

```
INPUT: evidence tensor e ∈ R^{B × K}, K=2 (HC, PD) OUTPUT: alpha, S, probabilities, uncertainty FOR each sample i in batch: alpha_k = e_k + 1.0 # Concentration parameters ≥ 1 S = sum(alpha_k, over k) # Total Dirichlet strength ≥ K p_k = alpha_k / S # Expected class probabilities u = K / S # Dirichlet vacuity ∈ (0, 1] RETURN alpha, S, p, u
```

B.2 Evidential Loss Computation (EvidentialLoss.forward)

```
INPUT: evidence e, target y (integer), epoch t, annealing_epochs t_max OUTPUT: scalar loss, loss components dict y_one_hot = one_hot(y, K) alpha = e + 1.0 S = sum(alpha, dim=-1) # Annealing coefficient lambda_t = min(1.0, t / t_max) # Expected Cross-Entropy (ACE) psi_S = digamma(S) psi_alpha = digamma(alpha) ace = sum(y_one_hot * (psi_S - psi_alpha), dim=-1) # per sample # KL Regularisation on Misleading Evidence alpha_tilde = y_one_hot + (1 - y_one_hot) * alpha kl = KL_Dirichlet(alpha_tilde, Dir(1)) # per sample total_loss = mean(ace) + lambda_t * mean(kl) RETURN total_loss, {ace, kl, lambda_t}
```

B.3 Cross-Modal Fusion Policy (5-Case Decision Logic)

```
INPUT: gait_output, voice_output, handwriting_output (any subset) OUTPUT: fused decision dictionary active_mods = [m for m in [gait, voice, handwriting] if m is not None] n = len(active_mods) IF n == 0: RAISE ValueError IF n == 1: RETURN Case E (single modality, inherited decision) fused_p_pd = uncertainty_weighted_average(active_mods) conformal_status = evaluate_conformal_agreement(active_mods) unanimous = all modalities agree on prediction class all_reliable = all mods ACCEPT and HIGH level all_unreliable = all mods FLAG and LOW level # Tri-Modal (n=3) or Dual-Modal (n=2): IF unanimous: IF all_reliable AND conformal_status == CONFORMAL_AGREEMENT: RETURN Case A (CONSISTENT, HIGH, ACCEPT) ELIF all_unreliable: RETURN Case D (INSUFFICIENT_EVIDENCE, LOW, FLAG) ELSE: RETURN Case B (PARTIAL_SUPPORT, MEDIUM, FLAG) ELSE: RETURN Case C (MODALITY_CONFLICT, LOW, FLAG)
```

B.4 Subject-Level Zero-Leakage Split

```
INPUT: manifest DataFrame with columns [subject_id, label, filepath] OUTPUT: train_df, val_df, test_df (disjoint subject sets) hc_subjects = unique subjects with label=0, sorted pd_subjects = unique subjects with label=1, sorted shuffle hc_subjects, pd_subjects with fixed seed n_train_hc, n_val_hc, n_test_hc = compute_counts(len(hc_subjects), 0.70, 0.15) # Similar for pd_subjects train_subjects = first n_train_hc HC + first n_train_pd PD val_subjects = next n_val_hc HC + next n_val_pd PD test_subjects = remaining HC + remaining PD # Rigorous assertions: ASSERT len(train ∩ val) == 0 # Zero leakage train-val ASSERT len(train ∩ test) == 0 # Zero leakage train-test ASSERT len(val ∩ test) == 0 # Zero leakage val-test RETURN manifest rows filtered by each subject set
```

Appendix C: Environment Setup and Execution Instructions

C.1 Prerequisites

- Python 3.9 or later
- pip package manager
- Minimum 8 GB RAM (16 GB recommended for Wav2Vec2)
- GPU optional but beneficial for Wav2Vec2 training
- Operating System: Windows, Linux, or macOS

C.2 Installation

```
# Clone repository git clone https://github.com/Mathesh07/parkinson-s-disease-detection.git cd parkinson-s-disease-detection # Create and activate virtual environment python -m venv venv # Linux/macOS: source venv/bin/activate # Windows: .\venv\Scripts\activate # Install dependencies pip install -r requirements.txt pip install -e . # Install as editable package
```

C.3 Dataset Placement

```
# Gait Dataset (PhysioNet) # Download from: https://physionet.org/content/gaitpdb/1.0.0/ # Place extracted files in: Gait/gait-in-parkinsons-disease-1.0.0/ # Voice Dataset (MDVR-KCL) # Place audio files in: Voice/26-29_09_2017_KCL/ReadText/{HC,PD}/*.wav Voice/26-29_09_2017_KCL/SpontaneousDialogue/{HC,PD}/*.wav # Handwriting Dataset (NewHandPD) # Place in: Handwriting/NewHandPD/HealthySpiral/, PatientSpiral/ Handwriting/NewHandPD/HealthyMeander/, PatientMeander/ Handwriting/NewHandPD/NewSpiral.csv, NewMeander.csv
```

C.4 Training Commands

```
# GAIT MODALITY python Gait/train.py # Standard baseline python Gait/train_evidential.py # Evidential two-stage training python Gait/evaluate_evidential.py # Test evaluation python -m Gait.experiments.leave_one_cohort_out # LOCO python -m Gait.experiments.deep_ensemble # M=5 ensemble python -m Gait.experiments.mc_dropout # MC Dropout python -m Gait.experiments.conformal_prediction # Conformal # VOICE MODALITY python Voice/train_evidential.py # Evidential Wav2Vec2 python Voice/evaluate_evidential.py # HANDWRITING MODALITY python Handwriting/train_evidential.py # Evidential ViT python Handwriting/evaluate_evidential.py
```

C.5 Inference Commands

```
# Unified tri-modal inference python predict_multimodal.py \ --gait Gait/sample_gait.txt \ --voice Voice/sample_audio.wav \ --handwriting Handwriting/sample_spiral.png # JSON output format python predict_multimodal.py --voice sample.wav --handwriting sample.png --json # Interactive Streamlit Demo Application streamlit run demo/app.py # Run all tests python -m pytest Gait/tests/ Voice/tests/ Handwriting/tests/ -v
```